

Computer Networks

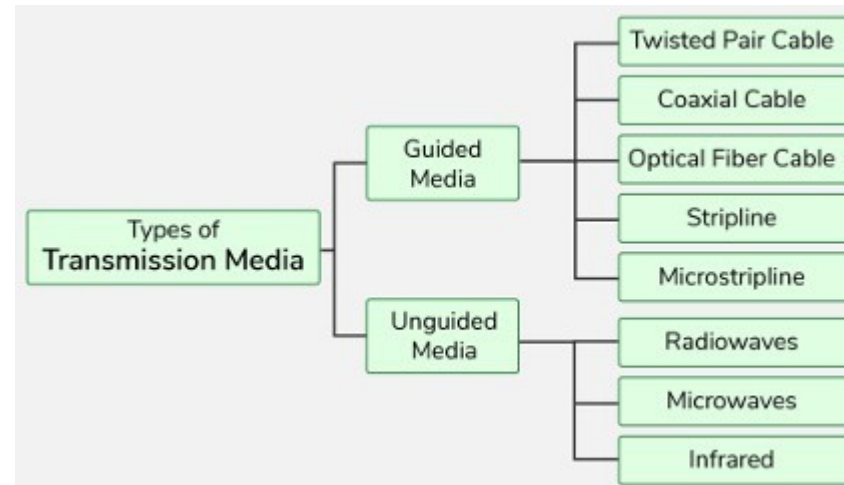
*[Types of Transmission
Media]*

Network Devices

- Transmission media is the physical medium through which data is transmitted from one device to another within a network. These media can be wired or wireless. The choice of medium depends on factors like distance, speed, and interference.

What is Transmission Media in Computer Networks?

- A transmission media is a physical path between the transmitter and the receiver i.e. it is the path along which data is sent from one device to another.
- Transmission Media is broadly classified into the following types.



Guided Media

- Guided Media is also referred to as Wired or Bounded transmission media.
- Signals being transmitted are directed and confined in a narrow pathway by using physical links.
- Features:
 - High Speed
 - Secure
 - Used for comparatively shorter distances

There are 3 major types of Guided Media:

- Twisted Pair Cable
- Coaxial Cable
- Optical Fiber Cable

Twisted Pair Cable

- It consists of 2 separately insulated conductor wires twisted about each other.
- Generally, several such pairs are bundled together in a protective sheath. They are the most widely used Transmission Media.
- Twisted Pair is of two types:
 - Unshielded Twisted Pair (UTP)
 - Shielded Twisted Pair (STP)

Unshielded Twisted Pair

(UTP)

- UTP consists of two insulated copper wires twisted around one another. This type of cable has the ability to block interference and does not depend on a physical shield for this purpose.
- It is used for telephonic applications.



Unshielded Twisted Pair

Advantages

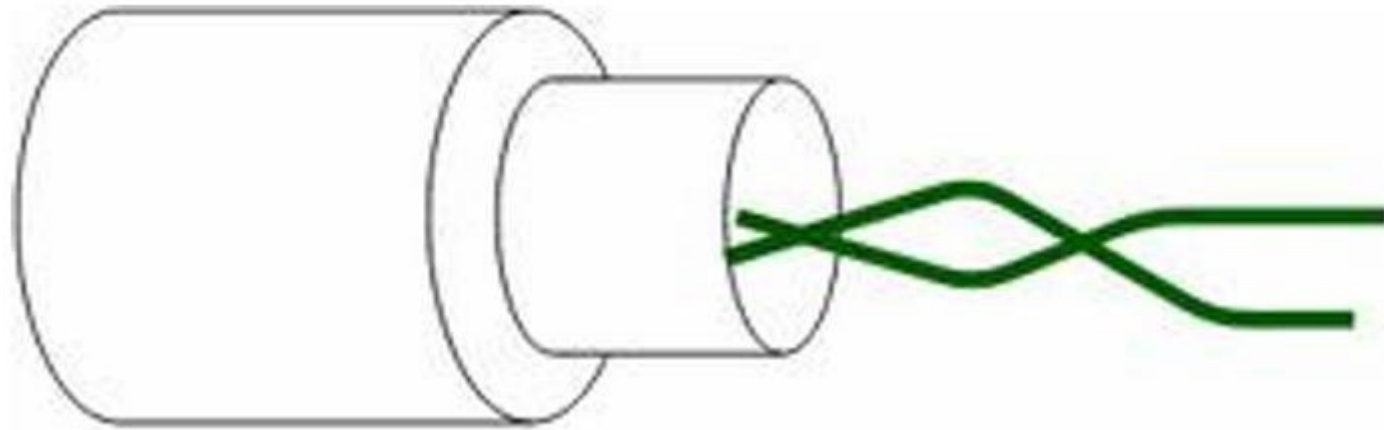
- Least expensive
- Easy to install
- High-speed capacity

Disadvantages

- Lower capacity and performance in comparison to STP
- Short distance transmission due to attenuation

Shielded Twisted Pair (STP)

- Shielded Twisted Pair (STP) cable consists of a special jacket (a copper braid covering or a foil shield) to block external interference.
- It is used in fast data rate Ethernet and in voice and data channels of telephone lines.



Shielded Twisted Pair

Advantages

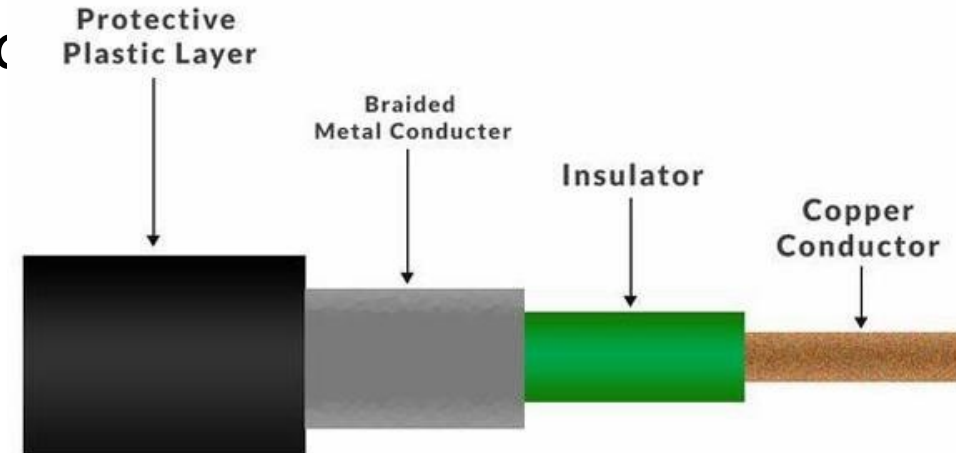
- Better performance at a higher data rate in comparison to UTP
- Eliminates crosstalk
- Comparatively faster

Disadvantages

- Comparatively difficult to install and manufacture
- More expensive
- Bulky

Coaxial Cable

- Coaxial cable has an outer plastic covering containing an insulation layer made of PVC or Teflon and 2 parallel conductors each having a separate insulated protection cover.
- The coaxial cable transmits information in two modes:
 - Baseband mode(dedicated cable bandwidth) and Broadband mode(cable bandwidth is split into separate ranges).
 - Cable TVs and analog television networks use



Coaxial Cable

Advantages

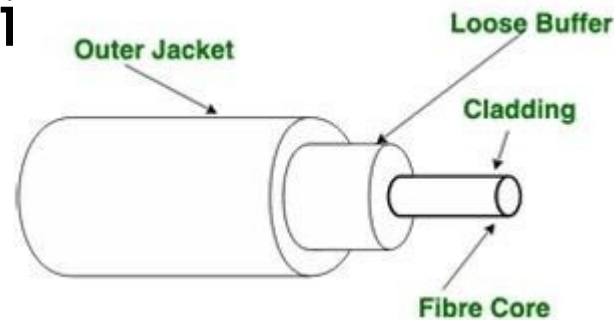
- Coaxial cables has high bandwidth .
- It is easy to install.
- Coaxial cables are more reliable and durable.
- Less affected by noise or cross-interference, electromagnetic.
- Coaxial cables support multiple channels.

Disadvantages

- Coaxial cables are expensive.
- The coaxial cable must be grounded in order to prevent any crosstalk.
- As a Coaxial cable has multiple layers, it is very difficult to tap.
- There is a chance of breaking the coaxial cable and attaching a "joint" by hackers, this compromises the security of the data.

Optical Fiber Cable

- Optical Fibre Cable uses the concept of total internal reflection of light through a core made up of glass.
- The core is surrounded by a less dense glass or plastic covering called the coating. It is used for the transmission of large volumes of data. The cable can be unidirectional or bidirectional.
- The WDM (Wavelength Division Multiplexer) supports two modes, namely unidirectional mode.



Optical Fiber Cable

Advantages

- Increased capacity and bandwidth
- Lightweight
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials

Disadvantages

- Difficult to install and maintain
- High cost

Applications of Optical Fibre Cable

- **Medical Purpose:** Used in several types of medical instruments.
- **Defence Purpose:** Used in transmission of data in aerospace.
- **For Communication:** This is largely used in formation of internet cables.
- **Industrial Purpose:** Used for lighting purposes and safety measures in designing the interior and exterior of automobiles.

Stripline

- Stripline is a transverse electromagnetic (TEM) transmission line medium invented by Robert M. Barrett of the Air Force Cambridge Research Centre in the 1950s.
- Stripline is the earliest form of the planar transmission line.
- It uses a conducting material to transmit high-frequency waves it is also called a waveguide.
- This conducting material is sandwiched between two layers of the ground plane which are usually shorted to provide EMI immunity.

Microstripline

- A microstripline is a type of transmission media used to carry high-frequency signals, commonly found in microwave and radio frequency circuits.
- It consists of a flat, narrow conducting strip (usually made of metal) placed on top of a dielectric material (an insulating layer), with a metal ground plane on the other side.

Unguided Media

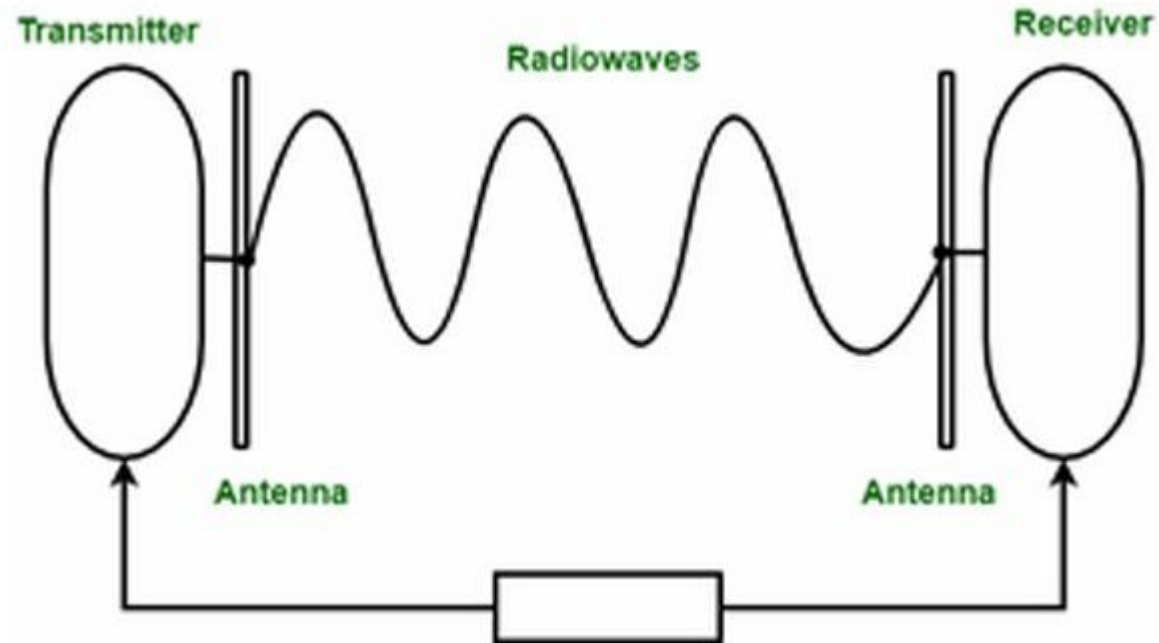
- It is also referred to as Wireless or Unbounded transmission media .
- No physical medium is required for the transmission of electromagnetic signals.
- Features of Unguided Media
 - The signal is broadcasted through air
 - Less Secure
 - Used for larger distances
- There are 3 types of Signals transmitted through unguided media:
 - Radiowaves
 - Microwaves
 - infrared

Radio Waves

- Radio waves are easy to generate and can penetrate through buildings. The sending and receiving antennas need not be aligned.
- **Frequency Range:** 3KHz - 1GHz.
- AM and FM radios and cordless phones use Radio waves for transmission.
 - Types of Radio Waves:
 - *Short Wave:* AM Radio
 - *VHF (Very High Frequency):* FM Radio/TV

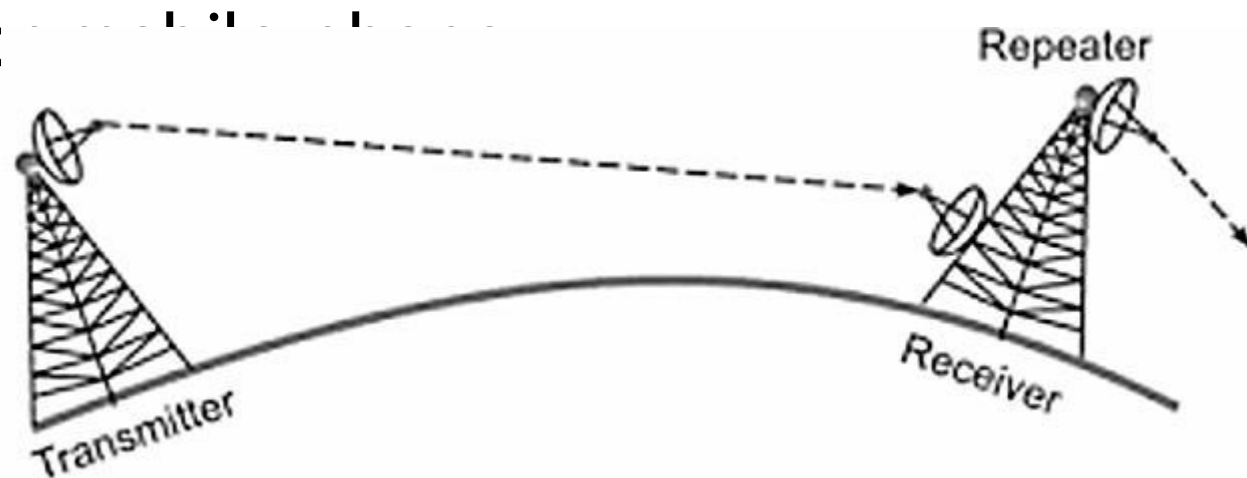
Radio Wave Components

- **Transmitter:** Responsible for encoding the signal.
- **Receiver:** Responsible for decoding the signal



Microwaves

- It is a line of sight transmission i.e. the sending and receiving antennas need to be properly aligned with each other.
- The distance covered by the signal is directly proportional to the height of the antenna.
- **Frequency Range:** 1GHz - 300GHz.
- Micro waves are majorly used for communication and television



Microwaves

Advantages

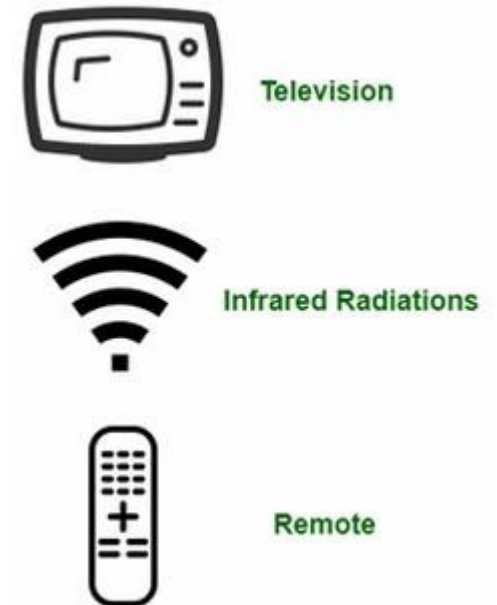
- Cheaper than using cables
- Freedom from land acquisition
- Ease of communication in difficult terrains
- Communication over oceans

Disadvantages

- Insecure communication.
- Out of phase signal.
- Susceptible to weather conditions.
- Bandwidth is limited.
- High cost of design, implementation, and maintenance.

Infrared

- Infrared waves are used for very short distance communication. They cannot penetrate through obstacles. This prevents interference between systems.
- **Frequency Range:** 300GHz - 400THz. It is used in TV remotes, wireless mouse, keyboard, printer, etc



Difference Between Radio Waves, Microwaves, and Infrared Waves

Basis	Radiowave	Microwave	Infrared Wave
Direction	These are omni directional in nature.	These are unidirectional in nature.	These are unidirectional in nature.
Penetration	At low frequency, they can penetrate through solid objects and walls but high frequency they bounce off the obstacle.	At low frequency, they can penetrate through solid objects and walls. at high frequency, they cannot penetrate.	They cannot penetrate through any solid object and walls.
Frequency range	Frequency range: 3 KHz to 1GHz.	Frequency range: 1 GHz to 300 GHz.	Frequency range: 300 GHz to 400 GHz.
Security	These offers poor security.	These offers medium security.	These offers high security.
Attenuation	Attenuation is high.	Attenuation is variable.	Attenuation is low.
Government License	Some frequencies in the radio-waves require government license to use these.	Some frequencies in the microwaves require government license to use these.	There is no need of government license to use these waves.
Usage Cost	Setup and usage Cost is moderate	Setup and usage Cost is high.	Usage Cost is very less
Communication	These are used in long distance communication.	These are used in long distance communication.	These are not used in long distance communication.

Causes of Transmission Impairment

- Transmission impairment refers to the loss or distortion of signals during data transmission, leading to errors or reduced quality in communication.
- Common causes include signal distortion, attenuation, and noise all of which can affect the clarity and reliability of transmitted data.

Causes of Transmission Impairment

- **Attenuation:** It means loss of energy. The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium. This is also known as attenuated signal.
- Amplifiers are used to amplify the attenuated signal which gives the original signal back and compensate for this loss.

Causes of Transmission Impairment

- **Distortion:** It means changes in the form or shape of the signal. This is generally seen in composite signals made up with different frequencies.
- Each frequency component has its own propagation speed travelling through a medium. And that's why it delays in arriving at the final destination. Every component arrives at different times which leads to distortion.
- Therefore, they have different phases at receiver end from what they had at sender's end.

Causes of Transmission Impairment

- **Noise:** The random or unwanted signal that mixes up with the original signal is called noise. There are several types of noise such as induced noise, crosstalk noise, thermal noise and impulse noise which may corrupt the signal.

Factors Considered for Designing the Transmission Media

- **Bandwidth:** Assuming all other conditions remain constant, the greater a medium's bandwidth, the faster a signal's data transmission rate.
- **Transmission Impairment :** Transmission Impairment occurs when the received signal differs from the transmitted signal. Signal quality will be impacted as a result of transmission impairment.
- **Interference:** Interference is defined as the process of disturbing a signal as it travels over a communication media with the addition of an undesired signal.

Applications of Transmission Media in Computer Networks

- Transmission media in computer networks are used to connect devices and transfer data. Here are some common applications:

Transmission Media	Application
Unshielded Twisted Pair (UTP)	Local Area Networks (LAN), telephones
Shielded Twisted Pair (STP)	Industrial networks, environments with high interference
Optical Fiber Cable	Long-distance communication, internet backbones
Coaxial Cable	Cable TV, broadband internet, CCTV
Stripline	Printed Circuit Boards (PCBs), microwave circuits
Microstripline	Antennas, satellite communication, RF circuits
Radio	Wireless communication, AM/FM radio, mobile phones
Infrared	Remote controls, short-range communication
Microwave	Satellite communication, radar, long-distance links